



PH 001: Preparatory Physics - 1

Course Code	PH 001
Course Name	Preparatory Physics - 1
L-T-P-C	3-1-0-4
Intended for	Preparatory Students
Distribution	Core
Approval	50th BoA (Jul 2023)

Course Contents

- **Units, Dimensions:** Basic dimensional analysis and a consolidated information of units of various physical observables. [3 Lectures]
- **Vector Analysis:** Vector multiplication and vector algebra, Vector analysis: gradient, divergence, and curl. Cartesian, cylindrical, plane polar and spherical polar coordinate system. [9 Lectures]
- **Concepts of Classical mechanics:** Velocity, linear momentum, acceleration, forced, Newton's law of motion, work and energy, conversion laws for energy and momentum, centre of mass, collision, moment of a force, angular momentum, conservation of angular momentum, moment of inertia. [12 Lectures]
- **Laws of Gravity:** Acceleration due to gravity, escape velocity, Kepler's laws of planetary motion. [6 Lectures]
- **Heat and Thermodynamics:** Thermal equilibrium, concept of temperature, thermometers and the Kelvin scale, heat, work, and internal energy. First law of thermodynamics. Second law of thermodynamics. [6 Lectures]
- **Waves and Acoustics:** Waves, propagating and stationary waves, interference of waves, sound wave. [6 Lectures]

Text Books

1. D. Halliday, R. Resnick, and J. Walker, **Fundamentals of Physics**, John Wiley.
2. Sears and Zemansky's **University Physics with Modern Physics**, 13th Edition, Pearson.